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ADAN 8888 - Applied Analytics Project - Spring 2026
Module 5 Assignment - Engineer Features

HelpHerInvest: A machine learning-based investment recommendation system that aligns user interests and risk preferences with benchmark stock evaluation.

Semantic Natural Language Processing

Synthetic User Inputs

Our team thought it would be beneficial to create synthetic user inputs to feed the natural language processing ("NLP") model for semantic analysis. The synthetic data is created by collecting randomizing topics across different categories and formatting them in a way that replicates a user's input. The categories range from technology, consumer, energy, healthcare, finance, etc. Within a given category we came up with several topics that are associated with that specific area. For example, within the technology category we linked topics such as AI, datacenters, cloud computing, cybersecurity, software, and quantum computing to name a few. We then have functions that randomize the topics collected and inserts them into templated inputs (i.e. "I want to invest in {topic}.") to eventually feed our NLP model. We constructed the function to even take in two topics within one user input to test the models performance for more complex prompts.

<u>Examples of Categories / Topics:</u>

```
topics = {
    "technology": ["AI", "cloud computing", "datacenters", "cybersecurity", "semiconductors","quantum computing","blockchain",
    "telecommunications","software","hardware","self-driving cars","automation"],
    "consumer": ["fashion", "gaming", "e-commerce", "beauty","home goods","food","beverage","travel","leisure","sports",
    "automotive","retail","consumer electronics"],
    "energy": ["renewable energy", "oil and gas", "EV infrastructure", "nuclear power","energy storage","solar power","wind
    energy","utilities","clean energy"],
    "healthcare": ["biotech", "pharmaceuticals", "medical devices","telemedicine","health IT","genomics","healthcare services",
    "wellness","fitness","mental health"],
    "media": ["music streaming", "broadcasting", "digital advertising","marketing","social media","entertainment","publishing",
    "gaming media","video streaming","podcasting"],
    "finance": ["fintech", "cryptocurrency", "banking", "insurance","wealth management","investment platforms","personal finance",
    "lending"],
}
```

Synthetic Text Inputs:

	primary_topic	secondary_topic	is_dual	text_input
0	cloud computing	utilities	1	I believe cloud computing and utilities will grow over the next decade.
1	telemedicine	None	0	I want to invest in telemedicine.
2	healthcare services	None	0	I'm passionate about healthcare services and want investment opportunities.
3	automation	None	0	I believe automation will grow significantly.
4	gaming	entertainment	1	I'm interested in both gaming and entertainment.
5	fintech	publishing	1	I want exposure to companies in fintech and publishing.
6	fintech	None	0	I'm passionate about fintech and want investment opportunities.
7	fitness	insurance	1	Looking to invest in fitness as well as insurance.
8	home goods	biotech	1	I believe home goods and biotech will grow over the next decade.
9	broadcasting	None	0	I'm interested in companies focused on broadcasting.
10	wealth management	None	0	I want to invest in wealth management.
11	personal finance	None	0	I'm passionate about personal finance and want investment opportunities.
12	insurance	None	0	Looking for exposure to insurance.
13	healthcare services	podcasting	1	I'm interested in both healthcare services and podcasting.
14	personal finance	None	0	I believe personal finance will grow significantly.
15	clean energy	None	0	Looking for exposure to clean energy.
16	cryptocurrency	digital advertising	1	I believe cryptocurrency and digital advertising will grow over the next decade.
17	lending	AI	1	Looking to invest in lending as well as AI.
18	podcasting	None	0	I'm passionate about podcasting and want investment opportunities.
19	leisure	insurance	1	I'm interested in both leisure and insurance.

Multinomial Classification Model

Feature Engineering

The financial dataset we have developed to make predictions on the forward excess return contains many variables that were built off existing features. For example, the momentum variables are statistics developed from calculating the change in ticker price from one period of time against another (i.e. 3-month, 6-month, 12-month). Similarly, the drawdown variables take the maximum ticker price at a point in time (6 months or 12 months prior) and compares it against the current ticker price. Each of the following variables is developed from feature engineering in some way:

- **1-Month Momentum**: The price percentage change over the last month
- **3-Month Momentum**: The price percentage change over the last 3 months
- **6-Month Momentum**: The price percentage change over the last 6 months
- **12-Month Momentum**: The price percentage change over the last 12 months
- **12-1-Month Momentum**: The price percentage change over the last 12 months, excluding the current month

- **3-Month Relative Strength Against S&P:** 3-month momentum difference from the S&P 3-month momentum
- **6-Month Relative Strength Against S&P:** 6-month momentum difference from the S&P 6-month momentum
- **12-Month Relative Strength Against S&P**: 12-month momentum difference from the S&P 12-month momentum

- **3-Month Volatility**: 3-month volatility calculation based on the standard deviation
- **6-Month Volatility**: 6-month volatility calculation based on the standard deviation

- **6-Month Drawdown**: 6-month comparison maximum stock price against current price
- **12-Month Drawdown:** 12-month comparison maximum stock price against current price

- **200-day rolling average:** 200-day rolling average of the stock price
- **Percentage above 200-day rolling average**: The current stock prices percentage above/below the 200-day rolling average price

```python
## Compute Features ##
monthly_px = prices.resample('ME').last()  # month-end prices
mom_1m  = monthly_px / monthly_px.shift(1)  - 1  # 1-month momentum
mom_3m  = monthly_px / monthly_px.shift(3)  - 1  # 3-month momentum
mom_6m  = monthly_px / monthly_px.shift(6)  - 1  # 6-month momentum
mom_12m = monthly_px / monthly_px.shift(12) - 1  # 12-month momentum
mom_12m_ex_1m = (monthly_px.shift(1) / monthly_px.shift(12)) - 1  # 12-month momentum excluding most recent month

rel_3m_spy  = mom_3m.sub(mom_3m["SPY"], axis=0)  # relative strength against S&P 3-month
rel_6m_spy  = mom_6m.sub(mom_6m["SPY"], axis=0)  # relative strength against S&P 6-month
rel_12m_spy = mom_12m.sub(mom_12m["SPY"], axis=0) # relative strength against S&P 12-month

daily_ret = prices.pct_change() # daily returns

vol_3m = (daily_ret.rolling(63).std() * np.sqrt(252)).resample("M").last() # 3-month volatility
vol_6m = (daily_ret.rolling(126).std() * np.sqrt(252)).resample("M").last() # 6-month volatility


roll_max_6m  = monthly_px.rolling(6).max()  # 6-month rolling max
roll_max_12m = monthly_px.rolling(12).max() # 12-month rolling max

drawdown_6m  = monthly_px / roll_max_6m  - 1  # 6-month drawdown
drawdown_12m = monthly_px / roll_max_12m - 1  # 12-month drawdown

dma_200 = prices.rolling(200).mean().resample("M").last() # 200-day moving average
pct_above_200dma = monthly_px / dma_200 - 1  # pct above 200-day moving average
```